**University of Minho**
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# Computational Discovery of DNA Aptamer-Graphene Sensors

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Master's Dissertation in Bioinformatics

Dissertation supervised by
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# Abstract

Rapid and reliable detection of microbial DNA is crucial in clinical diagnostics and environmental monitoring, but PCR-based methods, although highly sensitive, require complex instrumentation and workflows that limit their use at the point of care Hu et al. (2024). Graphene-based field-effect transistors (GFETs) offer an attractive alternative by converting local charge changes at the surface into measurable electrical signals, while DNA aptamers provide synthetic, programmable recognition elements with high affinity and specificity Krishnan et al. (2023); Gegenhuber et al. (2025); Sun et al. (2022). However, combining aptamers, microbial DNA targets, and graphene surfaces raises coupled challenges related to sequence specificity, structural stability under realistic conditions, Debye screening, nonspecific adsorption, and the interplay between folding, binding, and surface adsorption Green and Norton (2015); Purwidyantri et al. (2021).

This dissertation focuses on the computational characterization of DNA aptamer–microbial DNA complexes and their interactions with graphene, aiming to support the rational design of aptamer–graphene biosensors for microbial DNA detection Sun et al. (2022); Smith et al. (2020). Molecular modeling approaches, including docking, will be used to study binding modes, conformational dynamics, and adsorption behavior, in conjunction with published experimental data on graphene-based DNA sensors, particularly studies on electrolyte-gated graphene field-effect transistors and microfluidic graphene DNA sensors reported by Campos et al. (2019); Purwidyantri et al. (2022). The main goals are to identify sequence and structural features that support strong, selective binding compatible with graphene transduction, to assess how adsorption perturbs aptamer–target complexes, and to derive design principles for next-generation aptamer–graphene biosensors for health and environmental applications. These aims will be addressed using fully automated three-dimensional model generation and analysis workflows implemented in Python, relying on the GOPY module to build graphene-based systems and integrating docking for reproducible and scalable exploration of aptamer–DNA–graphene interactions. Hu et al. (2024); Krishnan et al. (2023).

# Chapter 1

# Introduction

## 1.1 Context

The fast identification of microbial DNA serves as the basis for infection diagnosis and environmental contamination tracking and antibiotic resistance monitoring. The detection limits of PCR-based methods and their associated amplification technologies are high, but these methods need dedicated equipment and laboratory environments with strict controlled conditions, and they require extended time for preparing samples Hu et al. (2024). The development of miniaturized solid-state biosensors now allows researchers to convert molecular recognition events into electrical or electrochemical signals which enables them to create small devices for portable and continuous monitoring applications Gegenhuber et al. (2025); Krishnan et al. (2023).

Combining DNA aptamers as recognition elements with graphene-based field-effect transistors has been proposed as a promising approach for label-free nucleic-acid detection Green and Norton (2015); Campos et al. (2019). The two-dimensional conduction channel of graphene detects local charge variations with high precision because aptamers function as DNA sequence detectors for microbial DNA sequences Sun et al. (2022); Hu et al. (2024). The device output reflects how aptamer structure and target complexes interact at the graphene surface, under ionic conditions that impose strong Debye screening and shape the effective sensing volume Green and Norton (2015); Purwidyantri et al. (2021). Building on experimental developments in graphene-based nucleic acid sensors, including electrolyte-gated graphene field-effect transistors and microfluidic graphene DNA devices reported by Campos et al. (2019); Purwidyantri et al. (2021, 2022), a key challenge is the high experimental cost and low throughput associated with testing hundreds of candidate aptamers to discriminate multiple pathogenic species or toxins, which motivates a complementary computational screening strategy.

## 1.2 Problem statement

The primary scientific challenge addressed in this dissertation is to computationally investigate, using three-dimensional model systems, how DNA aptamers designed to recognize specific microbial DNA sequences interact with their targets and subsequently with graphene surfaces. This research will rely on computational molecular modeling to analyze the factors, such as DNA sequence, microbial structure, and environmental conditions, that influence aptamer, microbial DNA binding affinity and specificity, as well as to determine how graphene adsorption modulates these interactions and charge distribution relevant for GFET-based sensing applications. Green and Norton (2015); Campos et al. (2019).

## 1.3 Objectives

This dissertation's main objective is to computationally characterize the binding between DNA aptamers and microbial DNA targets that are clinically relevant. It also explores the interactions between these complexes and graphene surfaces in configurations that are suitable for electrolyte-gated graphene field-effect transistor (GFET) biosensors.

This goal is decomposed into the following specific objectives:

- Select a panel of clinically relevant microbial targets and associated published probes, focusing on genes such as *lytA* and *ply* in *Streptococcus pneumoniae*, *nuc* and elongation factor *EF-TU* in *Staphylococcus aureus*, *ETA*, *oprL* and *algD* in *Pseudomonas aeruginosa*, and *frdB* in *Haemophilus influenzae*, based on experimental studies that report probes and targets for these genes, including graphene-based DNA sensor work by Campos et al. (2019); Purwidyantri et al. (2021, 2022).

- Build three-dimensional models of the selected DNA probes and their complementary microbial DNA sequences, and construct graphene slabs using the GOPY Python module, generating aptamer–target and aptamer–target–graphene systems suitable for docking.

- Perform molecular docking to characterize aptamer–target binding modes and predicted affinities, and to explore preferred orientations of aptamer–target complexes on graphene surfaces.

- Derive docking-based structural and simple electrostatic descriptors (e.g., contact patterns, interface areas, and charge distributions relative to the graphene plane) that can qualitatively inform GFET-relevant configurations.

- Compare the computationally obtained trends with experimental observations reported in graphene-based DNA sensor studies, in particular the work by Campos et al. (2019); Purwidyantri et al. (2021, 2022), with the aim of extracting practical design guidelines for aptamer–graphene biosensors.

## 1.4 Research questions

The research will center on key questions:

- What specific sequences and structural traits of DNA aptamers lead to effective and targeted binding with selected microbial DNA fragments, as quantified by docking scores and contact patterns from molecular dynamics simulations? Sun et al. (2022); Smith et al. (2020)

- How does adsorption onto graphene surfaces influence the conformation, binding strength, and interaction network of aptamer–DNA complexes, assessed through structural descriptors, interaction energies, and residence times at the graphene interface? Green and Norton (2015); Krishnan et al. (2023)

- Which geometrical arrangements and ionic conditions produce detectable shifts in aptamer-functionalized GFET response, inferred from simulated charge distributions, Debye layer characteristics, and proximity of key charges to the graphene plane? Campos et al. (2019); Purwidyantri et al. (2021)

- How can simulation-derived structural and energetic features be translated into practical design rules for aptamer–graphene sensors, using automated analysis and exploratory machine learning models to rank and prioritize aptamer candidates? Sun et al. (2022); Smith et al. (2020)

## 1.5 Methods (overview)

The proposed workflow begins with the construction of molecular models of functionalized DNA aptamers and representative microbial DNA sequences. This step employs Schrödinger suite tools, 3dDNA for secondary structure prediction, and AMBER for force field preparation, with automated model generation of the molecular systems with graphene surface built by the GOPY Python package to ensure reproducibility and scalability. Next, molecular docking and energy minimization procedures are used to simulate and refine aptamer–microbial DNA interactions, identifying stable binding conformations. Following this, the aptamer–microbial DNA complexes are evaluated in the context of graphene surfaces, enabling the analysis of adsorption behavior, orientation, and interaction energies. The workflow then systematically

compares the binding and non-binding states of aptamers with microbial DNA, examining their influence on aptamer–graphene interactions and biosensor signal transduction. All stages of model generation and analysis are automated using Python scripting and GOPY, supporting high-throughput investigation and reproducible research practices. Finally, machine learning models are integrated to predict and optimize structural and energetic properties of both aptamer–microbial DNA and aptamer–microbial DNA-graphene complexes. This enables adaptive biosensor design, facilitating the identification of sequence and structure features that enhance binding specificity and device performance. Sun et al. (2022); Smith et al. (2020); Krishnan et al. (2023).

# Chapter 2

# Tasks and planning

## 2.1 Task description

- **T1 – Model building** (Oct–Nov 2025): selecting microbial DNA segments and aptamer sequences; building three-dimensional structural models in solution; and generating graphene slab models using the GOPY Python module Sun et al. (2022); Green and Norton (2015).
- **T2 – Docking and binding states** (Dec 2025–Jan 2026): performing aptamer–microbial DNA docking to identify likely binding modes and prioritize candidates based on predicted binding affinity and interaction patterns Hu et al. (2024); Smith et al. (2020).
- **T3 – Aptamer–graphene docking** (Feb–Mar 2026): docking aptamer–DNA models onto graphene surfaces to analyze preferred orientations, contact interfaces, and qualitative indicators of transduction-relevant configurations Green and Norton (2015); Krishnan et al. (2023).
- **T4 – Workflow automation** (Apr–May 2026): developing Python scripts to automate model construction with GOPY, docking setup, and analysis, enabling systematic exploration of multiple sequences and conditions Sun et al. (2022); Campos et al. (2019).
- **T5 – Integration and advanced analysis** (Jun–Aug 2026): identifying promising sequence and structural motifs by integrating docking scores and interaction features and qualitatively mapping these metrics onto GFET-relevant design parameters Campos et al. (2019); Purwidyantri et al. (2021).
- **T6 – Writing and revision** (Sep 2026): organizing the dissertation, refining the state of the art, and discussing how docking-based insights support the design of aptamer–graphene biosensors Gegenhuber et al. (2025); Hu et al. (2024).

## 2.2 Timeline

| **Task** | Oct–Nov 25 | Dec 25–Jan 26 | Feb–Mar 26 | Apr–May 26 | Jun–Aug 26 | Sep 26 |
|---|---|---|---|---|---|---|
| T1: Model building | X | | | | | |
| T2: Aptamer–DNA docking | | X | | | | |
| T3: Aptamer–DNA–graphene docking | | | X | | | |
| T4: Automation | | | | X | | |
| T5: Integration and analysis | | | | | X | |
| T6: Writing and revision | | | | | | X |

## 2.3 Literature search methodology

In addition to Google Scholar and institutional repositories, the literature search will primarily utilize the Elicit AI platform and Consensus AI platform for the organized examination of recent work. PubMed will also be consulted. Search terms such as "DNA aptamer graphene," "graphene FET biosensor," "aptamer microbial detection," and "graphene DNA interactions" will be used as keywords, with a focus on current reviews and computational or experimental research directly related to aptamer–graphene sensors.

Elicit will be used to find complementary and less obvious references related to aptamer design and graphene biosensors, inclusion criteria will prioritize quality (journal reputation, citation count when available), recentness, and pertinence to the defined research questions.

## 2.4 Dissertation methodology

A computational workflow is used in order to assess aptamer-target binding and the interaction of these complexes with graphene surfaces in a manner that can be qualitatively compared with experimental graphene-based DNA sensors Campos et al. (2019); Purwidyantri et al. (2021, 2022).

1. **Target and probe selection.** A panel of clinically relevant microbial targets and associated DNA probes will be defined based on (i) species and genes reported in recent experimental studies on microbial diagnostics and graphene-based DNA sensors and (ii) published probes targeting genes such as *lytA* and *ply* in *Streptococcus pneumoniae*, *nuc* and elongation factor *EF-TU* in *Staphylococcus aureus*, *ETA*, *oprL* and *algD* in *Pseudomonas aeruginosa*, and *frdB* in *Haemophilus influenzae*. For each probe–target pair, the original experimental publication that reported and validated the probe will be recorded in a structured reference table.

2. **Three-dimensional model construction.** For each selected probe–target pair, three-dimensional structures of the DNA will be generated using nucleic-acid modeling tools, while graphene slabs will be constructed with the GOPY Python module to obtain atomistic models of the sensing interface. This yields a set of aptamer–target and aptamer–target–graphene systems suitable for docking calculations.

3. **Docking-based evaluation of aptamer–target binding.** Molecular docking will be used to characterize the binding between each DNA probe and its complementary microbial target sequence. For each pair, predicted binding modes, docking scores, and contact patterns will be analyzed to quantify sequence- and structure-dependent recognition. Aptamer–genomic DNA hybridization modeling uses NUPACK (web server) to predict ssDNA aptamer binding to 20–60 nt bacterial genomic fragments at 37°C and 0.1 M NaCl. Protocol: input aptamer and target sequences $\rightarrow$ compute $\Delta G$ (threshold $< -12$ kcal/mol), ensemble defect ($< 0.1$), and base-pairing probabilities ($> 0.8$); convert the dominant duplex (dot-bracket $\rightarrow$ forgi $\rightarrow$ 3DNA $\rightarrow$ AMBER minimization) to PDBQT format for AutoDock Vina graphene docking. The top five high-affinity duplexes ($\Delta G$ $-15$ to $-20$ kcal/mol) will feed the T3 workflow, enabling ssDNA versus dsDNA adsorption comparison for GFET signal prediction.

4. **Docking-based exploration of aptamer–target–graphene configurations.** Docking calculations will also be performed for aptamer–target models near graphene surfaces, to explore preferred orientations of the complexes at the interface and to identify configurations compatible with the formation of an effective sensing layer. Dock ssDNA aptamer models onto graphene using AutoDock Vina ($40 \times 40 \times 10$ grid), with PLIP/VMD analysis for $\pi$–$\pi$ stacking interfaces (>10 contacts, $\theta < 20°$ flatness), to predict GFET-relevant transduction configurations (flat ssDNA $\rightarrow$ n-doping; hybridization $\rightarrow$ desorption $\rightarrow$ $+\Delta V_{\text{Dirac}}$ shift).

5. **Descriptor analysis and qualitative mapping to GFET behaviour.** For each system, docking-

derived structural and simple electrostatic descriptors (such as contact areas, number and type of interfacial contacts, and distribution of charges relative to the graphene plane) will be summarized and compared across probes and targets. These computational features will be qualitatively mapped onto GFET-relevant design considerations, taking into account experimental findings from electrolyte-gated and back-gated graphene DNA sensors functionalized with DNA probes Campos et al. (2019); Purwidyantri et al. (2021, 2022); Zhan et al. (2016); Schedin et al. (2010).

6. **Comparison with experimental studies.** Trends identified in the docking analysis (for example, stronger predicted binding or more favorable graphene orientations) will be qualitatively compared with experimental performance reported for the same or similar probes and targets in graphene-based field-effect transistor devices and other DNA sensor studies reported in the literature. This comparison aims to evaluate the usefulness of the computational descriptors as indicators of sensor behaviour and to derive preliminary design guidelines for future aptamer–graphene biosensors Campos et al. (2019); Purwidyantri et al. (2021, 2022); Zhan et al. (2016); Schedin et al. (2010).

## 2.5 Expected outcomes

This project is expected to deliver:

- A curated list of clinically relevant microbial targets and published DNA probes, aligned with species and genes reported in recent experimental studies on microbial DNA detection and graphene-based DNA sensors.
- A set of three-dimensional models and docking-based characterizations of aptamer–target complexes and their orientations at graphene interfaces for multiple clinically relevant pathogens.
- Qualitative relationships between probe sequence and structural motifs and their predicted binding behaviour, both for aptamer–target duplexes and for aptamer–target complexes near graphene surfaces, under conditions compatible with GFET operation.
- A structured collection of experimental graphene-based DNA sensor studies linked to the corresponding computational models and descriptors.
- Design guidelines, supported by docking-derived descriptors and qualitative comparison with experimental studies, for aptamer–graphene sensor configurations targeting clinically relevant microbial DNA in the context of published graphene-based biosensor work.

# Chapter 3

# State of the art

DNA aptamers are short single-stranded nucleic acids selected by SELEX that fold into structures capable of high-affinity, high-specificity target recognition Stephens et al. (2025); Sun et al. (2022). The selection methods of Cell-SELEX and its related variants allow researchers to choose aptamers which bind to microbial cells and complicated targets thus producing aptamers that detect bacterial surface antigens and bacterial toxins and viral proteins and viral genomes Kolm et al. (2020); Hu et al. (2024). The chemical synthesis of aptamers provides better opportunities for solid-state biosensor development because these molecules can be easily modified and optimized through in silico methods according to Gegenhuber et al. (2025); Sun et al. (2022).

The detection method of aptamer-based biosensors depends on aptamer recognition which enables biological signals to produce optical or electrochemical or electrical responses Gegenhuber et al. (2025). "Aptamer platforms used for microbial DNA detection enable the identification of clinically relevant DNA targets through genomic or plasmid recognition, often combined with amplification or nanomaterial-based signal enhancement strategies. Graphene oxide FRET aptasensors, where single-stranded aptamers adsorb on GO (Graphene Oxide) and fluorescence is restored upon target-induced desorption, illustrate how aptamer folding and DNA binding can be coupled to carbon nanomaterial interfaces Zhu et al. (2015); Green and Norton (2015).

GFETs based on graphene use the material's high carrier mobility and its two-dimensional structure to detect surface events with excellent sensitivity according to Krishnan et al. (2023); Green and Norton (2015). The combination of electrolyte-gated GFETs with DNA probes allows scientists to detect hybridization attomolar levels through label-free measurements which shows potential for creating highly sensitive nucleic acid sensors Campos et al. (2019). The operating performance of devices depends on how their interfaces are designed and what conditions their electrolytes experience because high ionic strength solutions create Debye screening that minimizes distant charge effects which impact both signal and noise output Purwidyantri et al. (2021). The research by Purwidyantri et al. (2022) shows that microfluidics

technology allows graphene-based DNA sensors to function as part of programmable lab-on-chip systems.

These devices are based on how DNA and graphene interact with each other. Single-stranded DNA adheres robustly to graphene through $\pi$–$\pi$ stacking and associated interactions, while double-stranded DNA exhibits weaker binding due to base pairing and diminished nucleobase exposure. Green and Norton (2015); Krishnan et al. (2023). Adsorption can perturb DNA structure and kinetics, effects that are particularly important for aptamers, whose function depends on precise folding and conformational transitions Green and Norton (2015); Smith et al. (2020). In aptamer–graphene sensors, the aptamer must retain its recognition capability while interacting with graphene in a way that produces a useful electronic perturbation.

By connecting sequence and structure to binding and transduction, computational techniques aid in the design of aptamers and sensors Sun et al. (2022); Smith et al. (2020). Aptamer structures and binding modes with microbial DNA will be predicted using three-dimensional modeling and molecular docking, and preferred orientations of aptamer–DNA complexes on graphene, as well as the impact of ionic conditions, will be explored through docking-based configurations and simple electrostatic descriptors extracted from these models Green and Norton (2015); Krishnan et al. (2023). The rational design of aptamer–graphene systems will be supported by computational workflows that link docking-derived structural and charge-related features to GFET-relevant parameters, helping to qualitatively explain how changes in conformation and charge distribution can translate into electrical signals Campos et al. (2019); Purwidyantri et al. (2021).

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